Quantification: 548_hyp_HP_z20c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	17
interior lacuna count	14
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2786	1453	2384.39	px^2
major_axis_length_px	lacuna shape	84.5711	73.0575	38.5543	px
minor_axis_length_px	lacuna shape	42.6784	34.9392	19.6075	px
aspect_ratio	lacuna shape	2.0788	2.0264	0.6393	unitless
eccentricity	lacuna shape	0.8354	0.8698	0.1043	unitless
solidity	lacuna shape	0.782	0.7923	0.0867	unitless
roots_count	headline	11.0714	10.5	5.2691	count
ring_length_r30_px	headline	359.571	358.5	131.132	px
ring_length_r60_px	reported	674.643	641	235.885	px
owned_length_px	ownership-dependent	984.383	747.003	605.175	px
edge_count	ownership-dependent	53.1429	43.5	30.8242	count
mean_edge_length_px	ownership-dependent	18.3331	18.3546	3.06	px
perimeter_px	normalised	303.174	247.028	175.766	px
ring_area_r30_px2	normalised	9319	8093.5	3495.7	px^2
ring_area_r60_px2	normalised	20492.2	18114	6759.22	px^2
in_frame_fraction_r30	normalised	0.9836	1	0.0417	unitless
in_frame_fraction_r60	normalised	0.9528	1	0.0845	unitless
ring_density_r30	normalised	0.0389288	0.0391425	0.00475659	px^-1
ring_density_r60	normalised	0.0333249	0.0346516	0.00658469	px^-1
roots_per_100px_perimeter	normalised	4.0482	3.7708	1.5523	per 100 px
ring_attached_length_r30_px	network v2	323.429	320	133.673	px
ring_attached_length_r60_px	network v2	512.571	466	222.797	px
ring_length_w_r30_px	network v2	410.362	404.327	148.403	px
ring_length_w_r60_px	network v2	772.091	741.568	273.632	px
sholl_crossings_r10	network v2	8.6429	7.5	3.177	count
sholl_crossings_r20	network v2	8.5714	9	3.5673	count
sholl_crossings_r30	network v2	8.4286	8.5	2.6808	count
loop_count	loops, report only	2.7143	1.5	3.9307	count
loop_edge_share	loops, report only	0.1309	0.0871	0.1622	share
ring_width_mean_r30_px	width	6.9832	6.8926	0.7924	px

Per field
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measure	value	units
analysed_area_px2	1.00434e+06	px^2
total_skeleton_length_px	26424	px
canalicular_length_density_per_px	0.0263098	px^-1
junction_count	566	count
junction_density_per_px2	0.00056355	px^-2
lacuna_count	17	count

lacunae_per_px2	1.693e-05	px^-2
skeleton_component_count	424	count
mean_component_length_px	62.3208	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0299697	px^-1
field_density_without_flagged_per_px	0.0252485	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	463	px
unattached_component_count	25	count
canalicular_length_density_attached_per_px	0.0263098	px^-1
field_length_density_with_unattached_per_px	0.0267708	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	56	count
loop_edge_share	0.1394	share
edge_count	1535	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.3141	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	22844	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	983	77.6846	19.2927	4.0266	0.9687	0.7097	-0.7276	26.3245	837.443	True
2	1345	60.6773	30.0142	2.0216	0.8691	0.8739	0.6556	34.9123	469.557	False
3	5304	115.818	61.0253	1.8979	0.8499	0.8497	-0.5279	76.0664	899.181	False
4	1100	56.8071	27.1347	2.0935	0.8785	0.8035	0.1617	61.0436	700.865	False
5	589	35.3245	25.9374	1.3619	0.6789	0.8158	0.6135	217.115	155.742	False
6	1020	65.1958	29.2967	2.2254	0.8933	0.5543	0.7245	258.288	921.093	False
7	8181	176.425	74.8525	2.357	0.9055	0.7394	-0.9469	333.681	384.601	False
8	772	43.3165	23.8026	1.8198	0.8355	0.8863	1.0039	324.626	799.803	False
9	843	67.1384	19.7863	3.3932	0.9556	0.7938	-0.8751	303.813	882.996	False
10	3628	78.9766	63.2544	1.2486	0.5988	0.8761	0.6151	443.631	464.428	False
11	1526	56.8255	39.8642	1.4255	0.7126	0.7592	0.9788	480.14	672.478	False
12	2634	99.9739	44.742	2.2345	0.8943	0.697	1.1806	489.293	203.02	False
13	5796	118.655	71.2141	1.6662	0.7999	0.7445	0.2563	543.606	765.066	False
14	2266	83.3012	36.8142	2.2627	0.897	0.8389	0.7457	720.831	996.501	True
15	1380	84.7713	25.484	3.3264	0.9537	0.7908	-1.481	787.319	890.853	False
16	1983	87.7473	31.9283	2.7483	0.9315	0.8015	-1.3962	877.209	988.193	True
17	4886	124.09	61.0897	2.0313	0.8704	0.7632	0.9159	965.955	722.24	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	10	265	443	586.558	42	13.9657	6.1708
2	5	254	605	540.002	26	20.7693	7.0371
3	9	376	575	751.896	36	20.886	8.3236
4	10	326	577	1230.09	58	21.2084	7.0894
5	9	252	559	1161.54	48	24.1988	6.4411
6	13	341	675	568.772	30	18.9591	6.5011
7	26	699	1239	1727.31	97	17.8074	6.6292
8	3	181	250	337.919	18	18.7733	7.9462
9	9	203	409	349.12	25	13.9648	5.5052
10	12	376	759	609.345	39	15.6242	6.7482
11	14	299	607	2302.77	110	20.9342	7.1631
12	12	460	949	1381.07	77	17.936	7.1896
13	13	450	842	742.11	51	14.5512	8.2997
14	5	276	454	300.309	18	16.6838	7.0874
15	9	393	694	1660.46	100	16.6046	6.7155
16	6	290	592	786.039	40	19.651	7.3308
17	11	424	705	418.948	29	14.4465	6.1752

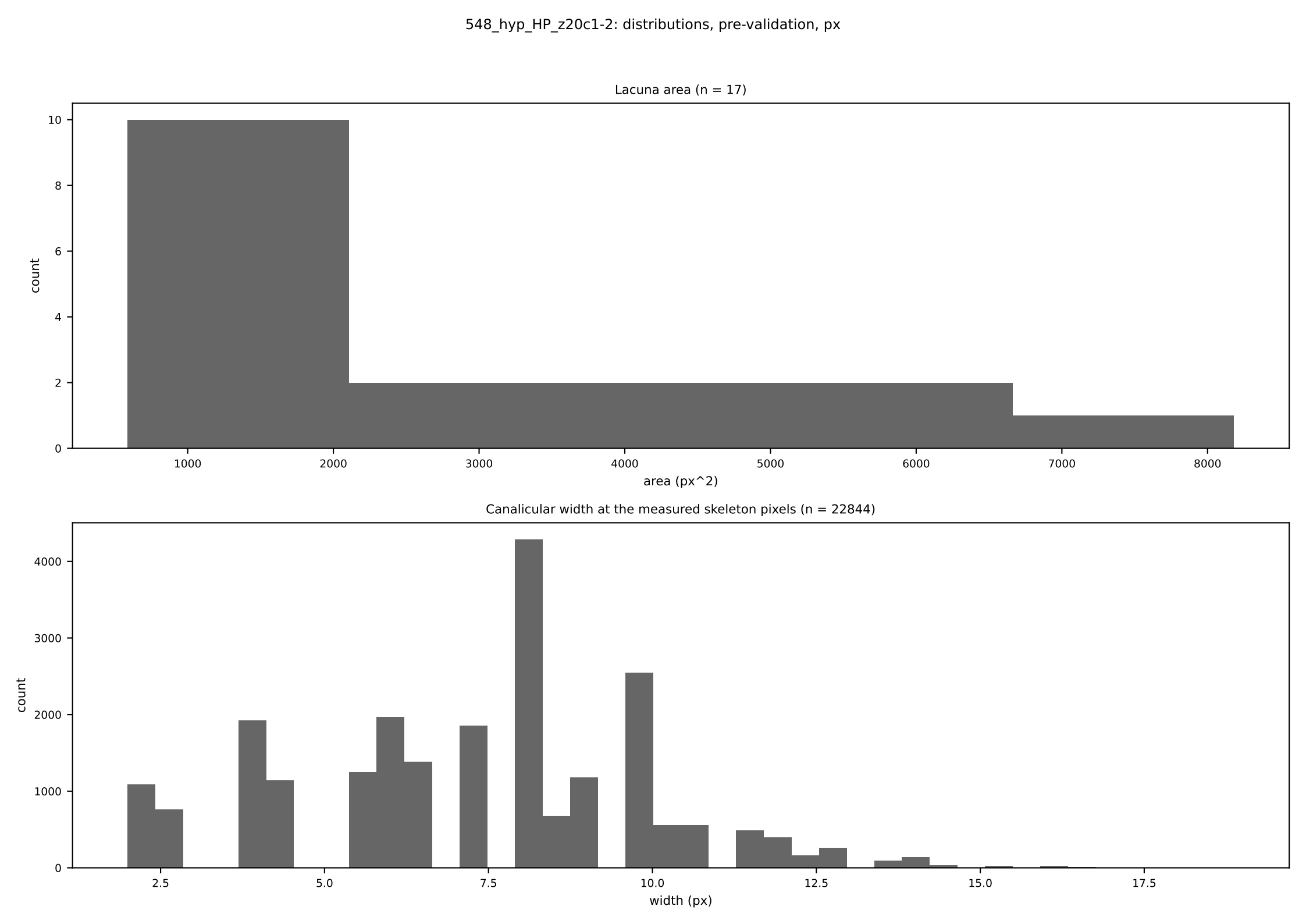
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	191.238	5199	8502	0.7613	0.6818	0.0509713	0.0521054	5.2291
2	185.238	6586	15515	0.8654	0.7462	0.0385667	0.0389945	2.6992
3	355.505	11178	21571	0.996	0.8914	0.0336375	0.0266562	2.5316
4	180.894	7279	17216	1	0.8909	0.0447864	0.0335153	5.5281
5	130.368	5996	17558	1	1	0.042028	0.0318373	6.9036
6	247.806	7577	17555	1	1	0.0450046	0.0384506	5.246
7	732.927	17463	35109	1	1	0.0400275	0.0352901	3.5474
8	129.095	5795	14290	1	1	0.0312338	0.0174947	2.3239
9	153.238	5367	9246	1	1	0.0378237	0.0442353	5.8732
10	300.433	10348	22315	1	1	0.0363355	0.034013	3.9942
11	246.25	7528	16500	1	1	0.0397184	0.0367879	5.6853
12	351.747	10779	26637	1	1	0.0426756	0.0356271	3.4115
13	533.985	13882	29213	1	1	0.0324161	0.0288228	2.4345
14	256.108	6631	13116	0.7153	0.654	0.0416227	0.0346142	1.9523
15	213.137	8610	18670	1	1	0.0456446	0.0371719	4.2226
16	256.551	6858	14534	0.748	0.6875	0.0422864	0.0407321	2.3387
17	483.813	12078	25496	0.909	0.8104	0.0351052	0.0276514	2.2736

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	235	346	301.279	510.031	7	3	6
2	199	297	290.693	670.103	6	6	7
3	328	506	423.676	649.428	8	13	9
4	287	441	370.22	647.558	7	9	7
5	200	399	285.451	638.257	7	6	8
6	323	425	390.848	780.867	9	8	8
7	671	1060	797.826	1438.41	17	16	12
8	151	185	211.137	285.664	6	4	5
9	166	328	237.208	482.588	5	2	3
10	317	491	417.806	848.24	7	9	12
11	299	547	346.392	702.269	6	7	6
12	393	694	519.517	1086.36	11	9	11
13	439	776	517.76	967.306	10	11	10
14	146	237	305.794	508.718	5	6	9
15	388	632	460.961	815.122	11	10	11

16	256	545	327.936	679.642	8	6	11
17	367	395	475.576	797.11	11	10	9


548_hyp_HP_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 17)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22844)
count
width (px)